Midterm - Representations of Quivers

April 13th, 2026

Choose up to five problems from this list to submit.

1. Let Q be the quiver $1 \longleftarrow 2 \longrightarrow 3$ and let M be a representation

$$k \xleftarrow{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}} k^3 \xrightarrow{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}} k^2 .$$

Write M as a direct sum of simple representations, indecomposable projective and injective representations.

Proof. Identify Q with $1 \xleftarrow{\alpha} 2 \xrightarrow{\beta} 3$. Then M is given by

$$M_1 = k, \qquad M_2 = k^3, \qquad M_3 = k^2,$$

with (column vectors) $\phi_\alpha = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} : k^3 \to k$ and $\phi_\beta = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} : k^3 \to k^2$.

Step 1: explicit summands. Let e_1, e_2, e_3 be the standard basis of k^3 and f_1, f_2 the standard basis of k^2. Define subrepresentations $M^{(1)}, M^{(2)}, M^{(3)} \subseteq M$ by

$$M_1^{(1)} = k, \quad M_2^{(1)} = \langle e_1 \rangle, \quad M_3^{(1)} = \langle f_1 \rangle,$$

$$M_1^{(2)} = 0, \quad M_2^{(2)} = \langle e_2 \rangle, \quad M_3^{(2)} = \langle f_2 \rangle,$$

$$M_1^{(3)} = 0, \quad M_2^{(3)} = \langle e_3 \rangle, \quad M_3^{(3)} = 0.$$

The restrictions of ϕ_α, ϕ_β are well-defined on each summand, and at every vertex the subspaces sum to M_1, M_2, M_3. Hence $M = M^{(1)} \oplus M^{(2)} \oplus M^{(3)}$.

Step 2: identify each piece.

- $M^{(3)}$ has dimension vector $(0,1,0)$ and both arrow maps are zero; thus $M^{(3)} \cong S(2)$ (simple).
- $M^{(1)}$ has dimension vector $(1,1,1)$; the restrictions of ϕ_α and ϕ_β are isomorphisms $\langle e_1 \rangle \to k$ and $\langle e_1 \rangle \to \langle f_1 \rangle$. This is the indecomposable projective $P(2)$: indeed $P(2)_j$ has a basis of paths $2 \to j$, namely e_2, α, β, and both structure maps send the generator of $P(2)_2$ to the corresponding basis vectors at 1 and 3.
- $M^{(2)}$ has dimension vector $(0,1,1)$; $\phi_\alpha = 0$ and ϕ_β identifies $\langle e_2 \rangle$ with $\langle f_2 \rangle$. This is the indecomposable injective $I(3)$: paths ending at 3 are e_3 and β from 2, so $I(3)_2$ and $I(3)_3$ are one-dimensional and the map along β is the identity in the path basis (same picture as $I(2)$ in the notes for $1 \to 2 \leftarrow 3$, but supported on the edge $2 \to 3$).

Conclusion.

$$M \;\cong\; P(2) \oplus I(3) \oplus S(2),$$

a direct sum of an indecomposable projective, an indecomposable injective, and a simple representation. ∎

2. Let Q be the quiver

$$1 \underset{\beta}{\overset{\alpha}{\leftleftarrows}} 2$$

For any $\lambda \in k \cup \{\infty\}$, define M_λ to be the representation:

$$1 \underset{\lambda}{\overset{1}{\leftleftarrows}} 2 \qquad \lambda \in k;$$

$$1 \underset{1}{\overset{0}{\leftleftarrows}} 2 \qquad \lambda = \infty.$$

(a) Show that each M_λ is indecomposable.
(b) Show that $M_\lambda \simeq M_\mu$ if and only if $\lambda = \mu$. In particular, the number of indecomposable representations depends on the choice of the field k.
(c) Show that $\operatorname{Hom}(M_\lambda, M_\mu) = 0$ if $\lambda \neq \mu$.

(a) *Proof.* Each M_λ has dimension vector

$$\underline{\dim}\, M_\lambda = (1, 1).$$

Suppose, for contradiction, that

$$M_\lambda \cong N_1 \oplus N_2$$

is a nontrivial decomposition with $N_1, N_2 \neq 0$. Since

$$\underline{\dim}\, N_1 + \underline{\dim}\, N_2 = (1, 1),$$

one of the two summands must have dimension vector $(0, 1)$. Without loss of generality, assume

$$\underline{\dim}\, N_2 = (0, 1).$$

Then $N_2(1) = 0$ and $N_2(2) = k$, so N_2 would have to be a subrepresentation of M_λ. Since both arrows α, β go from vertex 2 to vertex 1, this would force

$$M_\lambda(\alpha)(k) \subseteq N_2(1) = 0, \qquad M_\lambda(\beta)(k) \subseteq N_2(1) = 0.$$

If $\lambda \in k$, then

$$M_\lambda(\alpha) = 1\colon k \to k,$$

so

$$M_\lambda(\alpha)(k) = k \neq 0,$$

a contradiction.
If $\lambda = \infty$, then

$$M_\infty(\beta) = 1\colon k \to k,$$

so

$$M_\infty(\beta)(k) = k \neq 0,$$

again a contradiction.
Therefore no such nontrivial decomposition exists, and M_λ is indecomposable. $\square$

(b) *Proof.* ($\Leftarrow$) If $\lambda = \mu$, then the identity maps on the two vertex spaces give an isomorphism $M_\lambda \cong M_\mu$.

($\Rightarrow$) Suppose that
$$f = (f_1, f_2)\colon M_\lambda \to M_\mu$$
is an isomorphism. Since both vertex spaces are k, we may write
$$f_1 = a \in k^\times, \qquad f_2 = b \in k^\times.$$
The commutativity conditions for the arrows $\alpha, \beta\colon 2 \to 1$ are
$$f_1 \circ M_\lambda(\alpha) = M_\mu(\alpha) \circ f_2, \qquad f_1 \circ M_\lambda(\beta) = M_\mu(\beta) \circ f_2.$$
Case 1: $\lambda, \mu \in k$.
Then
$$M_\lambda(\alpha) = 1, \qquad M_\lambda(\beta) = \lambda, \qquad M_\mu(\alpha) = 1, \qquad M_\mu(\beta) = \mu.$$
Hence
$$a \cdot 1 = 1 \cdot b, \qquad a\lambda = \mu b.$$
From the first equation we get $a = b$, and substituting into the second gives
$$a\lambda = \mu a.$$
Since $a \neq 0$, we conclude that $\lambda = \mu$.
Case 2: $\lambda \in k$ and $\mu = \infty$.
Then
$$M_\lambda(\alpha) = 1, \qquad M_\infty(\alpha) = 0.$$
So commutativity at α gives
$$a \cdot 1 = 0 \cdot b = 0,$$
hence $a = 0$, contradicting the fact that f_1 is invertible.
Case 3: $\lambda = \infty$ and $\mu \in k$.
Then
$$M_\infty(\alpha) = 0, \qquad M_\mu(\alpha) = 1.$$
So commutativity at α gives
$$a \cdot 0 = 1 \cdot b,$$
hence $b = 0$, contradicting the fact that f_2 is invertible.
Thus the only possible case is $\lambda = \mu$. Therefore
$$M_\lambda \cong M_\mu \iff \lambda = \mu.$$
In particular, the family $\{M_\lambda\}_{\lambda \in k \cup \{\infty\}}$ contains $|k| + 1$ pairwise non-isomorphic indecomposable representations, so the number of indecomposable representations depends on the field k. $\square$

(c) *Proof.* Let
$$f = (f_1, f_2)\colon M_\lambda \to M_\mu$$
be a morphism. Since both vertex spaces are k, write
$$f_1 = a \in k, \qquad f_2 = b \in k.$$
Again, the commutativity conditions are
$$f_1 \circ M_\lambda(\alpha) = M_\mu(\alpha) \circ f_2, \qquad f_1 \circ M_\lambda(\beta) = M_\mu(\beta) \circ f_2.$$

Case 1: $\lambda, \mu \in k$ and $\lambda \neq \mu$.
Then
$$M_\lambda(\alpha) = 1, \qquad M_\lambda(\beta) = \lambda, \qquad M_\mu(\alpha) = 1, \qquad M_\mu(\beta) = \mu.$$
Hence
$$a = b, \qquad a\lambda = \mu b.$$
Substituting $b = a$ into the second equation gives
$$a(\lambda - \mu) = 0.$$
Since $\lambda \neq \mu$, we get $a = 0$, and hence $b = 0$.
Case 2: $\lambda \in k$ and $\mu = \infty$.
Then
$$M_\lambda(\alpha) = 1, \qquad M_\infty(\alpha) = 0,$$
so the condition at α gives
$$a \cdot 1 = 0 \cdot b = 0,$$
hence $a = 0$. Now the condition at β becomes
$$0 \cdot \lambda = 1 \cdot b,$$
so $b = 0$.
Case 3: $\lambda = \infty$ and $\mu \in k$.
Then
$$M_\infty(\alpha) = 0, \qquad M_\mu(\alpha) = 1,$$
so the condition at α gives
$$a \cdot 0 = 1 \cdot b,$$
hence $b = 0$. Then the condition at β becomes
$$a \cdot 1 = \mu \cdot 0 = 0,$$
so $a = 0$.
In every case with $\lambda \neq \mu$, we obtain $a = b = 0$. Therefore
$$\mathrm{Hom}(M_\lambda, M_\mu) = 0.$$
□

3. Let Q be a quiver, $L \xrightarrow{f} M \xrightarrow{g} N$ be an exact sequence in $rep\,Q$, f is injective, and X be representation of Q.
   Show that $\mathrm{Ker}\, g_* \subseteq \mathrm{Im}\, f_*$, where f_* and g_* are defined in
$$\mathrm{Hom}(X, L) \xrightarrow{f_*} \mathrm{Hom}(X, M) \xrightarrow{g_*} \mathrm{Hom}(X, N)\ .$$

   Proof. Let
$$h \in \mathrm{Ker}\, g_*.$$
   By definition of g_*, this means
$$g_*(h) = 0,$$
   so
$$g \circ h = 0.$$

Since

$$L \xrightarrow{f} M \xrightarrow{g} N$$

is exact in $\operatorname{rep} Q$, we have

$$\operatorname{Im} f = \operatorname{Ker} g.$$

Equivalently, $f\colon L \to M$ is a kernel of g.

Now $h\colon X \to M$ satisfies $g \circ h = 0$. Therefore, by the universal property of the kernel, there exists a morphism

$$u\colon X \to L$$

such that

$$f \circ u = h.$$

But then, by definition of f_*,

$$f_*(u) = f \circ u = h.$$

Hence

$$h \in \operatorname{Im} f_*.$$

Since $h \in \operatorname{Ker} g_*$ was arbitrary, it follows that

$$\operatorname{Ker} g_* \subseteq \operatorname{Im} f_*.$$

□

4. Let $g : L \to M$ be an injective morphism between representations of Q, and let $I(i)$ be the injective representation at vertex i. Show that the map

$$g^* : \operatorname{Hom}(M, I(i)) \to \operatorname{Hom}(L, I(i))$$

is surjective.

Proof. For every representation X, there is a natural isomorphism

$$\Psi_X\colon \operatorname{Hom}(X, I(i)) \xrightarrow{\sim} X_i^*,$$

where $X_i^* = \operatorname{Hom}_k(X_i, k)$. Concretely, after identifying

$$I(i)_i \cong k$$

via the trivial path e_i, a morphism $f\colon X \to I(i)$ is sent to the linear functional

$$\Psi_X(f) = f_i \in \operatorname{Hom}_k(X_i, k) = X_i^*.$$

Under these identifications, the map

$$g^*\colon \operatorname{Hom}(M, I(i)) \to \operatorname{Hom}(L, I(i)), \qquad \psi \mapsto \psi \circ g,$$

corresponds to the dual map

$$g_i^*\colon M_i^* \to L_i^*, \qquad \lambda \mapsto \lambda \circ g_i.$$

Indeed, for every $\psi \in \operatorname{Hom}(M, I(i))$, we have

$$\Psi_L\big(g^*(\psi)\big) = \Psi_L(\psi \circ g) = (\psi \circ g)_i = \psi_i \circ g_i = g_i^*(\psi_i) = g_i^*\big(\Psi_M(\psi)\big).$$

Hence the following diagram commutes:

$$\begin{tikzcd} \mathrm{Hom}(M, I(i)) \arrow[r, "\Psi_M", "\sim"'] \arrow[d, "g^*"'] & M_i^* \arrow[d, "g_i^*"] \\ \mathrm{Hom}(L, I(i)) \arrow[r, "\Psi_L", "\sim"'] & L_i^* \end{tikzcd}$$

Now $g\colon L \to M$ is injective, so each component

$$g_j\colon L_j \to M_j$$

is injective; in particular, g_i is injective. Since we are working with finite-dimensional representations, L_i and M_i are finite-dimensional vector spaces. Therefore the dual map

$$g_i^*\colon M_i^* \to L_i^*$$

is surjective.

For completeness, let $\lambda \in L_i^*$. Since g_i is injective, it induces an isomorphism

$$L_i \xrightarrow{\sim} g_i(L_i) \subseteq M_i.$$

Thus λ defines a linear form on the subspace $g_i(L_i)$ by

$$g_i(x) \longmapsto \lambda(x).$$

By linear algebra, this linear form extends to some $\mu \in M_i^*$. Then

$$g_i^*(\mu) = \mu \circ g_i = \lambda.$$

Hence g_i^* is surjective.

Since Ψ_M and Ψ_L are isomorphisms and the diagram above commutes, it follows that

$$g^*\colon \mathrm{Hom}(M, I(i)) \to \mathrm{Hom}(L, I(i))$$

is surjective. $\square$

5. Prove that any projective representation corresponding to vertex i, $P(i)$ is indecomposable.

Proof. Let Q be acyclic and $P(i) = (P(i)_j, \varphi_\alpha)$ the projective representation at i. Then $P(i)_i \cong k$ (spanned by the lazy path e_i).

Suppose $P(i) = M \oplus N$ for some $M, N \in \mathrm{Rep}\, Q$. After reordering, we may assume $P(i)_i = M_i$ and $N_i = 0$.

If $N \neq 0$, choose a vertex ℓ with $N_\ell \neq 0$. The space $P(i)_\ell$ has a basis consisting of all paths $c = (i \mid \beta_1, \ldots, \beta_s \mid \ell)$ from i to ℓ. Let $\varphi_c = \varphi_{\beta_s} \cdots \varphi_{\beta_1}$ be the composition of the structure maps of $P(i)$ along c. Because $P(i) = M \oplus N$, the map φ_c sends $M_i \oplus 0$ into $M_\ell \oplus N_\ell$. But $\varphi_c(e_i) = c$, so the basis element $c \in P(i)_\ell$ lies in M_ℓ. Hence every basis path $c \in P(i)_\ell$ lies in M_ℓ, contradicting $N_\ell \neq 0$.

Therefore $N = 0$ and $P(i)$ is indecomposable. $\square$

6. Compute a projective resolutions for representations $S(1)$ and $S(2)$ for quiver

$$Q: \quad \begin{tikzcd} 1 \arrow[r] \arrow[rr, bend left] & 2 \arrow[r] \arrow[d] & 3 \arrow[dl] \\ & 4 & \end{tikzcd}$$

Proof. Recall that for each simple representation $S(i)$, its minimal projective resolution is obtained from the projective cover

$$P(i) \twoheadrightarrow S(i),$$

whose kernel is $\operatorname{rad} P(i)$.

Moreover, for a path algebra of an acyclic quiver, one has

$$\operatorname{rad} P(i) \cong \bigoplus_{\alpha: i \to j} P(j),$$

where the direct sum runs over all arrows starting at i.

Resolution of $S(2)$. There are two arrows starting at 2, namely

$$\gamma : 2 \to 3, \qquad \delta : 2 \to 4.$$

Hence

$$\operatorname{rad} P(2) \cong P(3) \oplus P(4).$$

Therefore

$$0 \longrightarrow P(3) \oplus P(4) \xrightarrow{h} P(2) \xrightarrow{\pi_2} S(2) \longrightarrow 0$$

is exact, where $h = (\iota_\gamma, \iota_\delta)$ and:

- $\iota_\gamma : P(3) \to P(2)$ sends the generator of $P(3)_3 \cong k$ to
$$\gamma \in P(2)_3 \cong k;$$
- $\iota_\delta : P(4) \to P(2)$ sends the generator of $P(4)_4 \cong k$ to
$$\delta \in P(2)_4 \cong k^2.$$

The image of h is exactly the radical of $P(2)$, so this is the minimal projective resolution of $S(2)$.

Resolution of $S(1)$. There are two arrows starting at 1, namely

$$\alpha : 1 \to 2, \qquad \beta : 1 \to 3.$$

Thus

$$\operatorname{rad} P(1) \cong P(2) \oplus P(3).$$

Hence we have the exact sequence

$$0 \longrightarrow P(2) \oplus P(3) \xrightarrow{u} P(1) \xrightarrow{\pi_1} S(1) \longrightarrow 0,$$

where $u = (\iota_\alpha, \iota_\beta)$ and:

- $\iota_\alpha : P(2) \to P(1)$ sends the generator of $P(2)_2 \cong k$ to
$$\alpha \in P(1)_2 \cong k;$$
- $\iota_\beta : P(3) \to P(1)$ sends the generator of $P(3)_3 \cong k$ to
$$\beta \in P(1)_3 \cong k^2.$$

Its image is $\operatorname{rad} P(1)$, so this yields the minimal projective resolution of $S(1)$. □

7. Let $M = (M_i, \psi_\alpha)$ be a representation of Q. Prove that for any vertex i in Q, there is an isomorphism of vector spaces:

$$\operatorname{Hom}(M, I(i)) \simeq M_i.$$

Proof. Recall that $I(i)_j$ has as a basis the set of all paths from j to i in Q, and that for an arrow $\alpha\colon j \to \ell$, the structure map

$$I(i)_\alpha\colon I(i)_j \to I(i)_\ell$$

is defined on a basis element (a path $p\colon j \to i$) by

$$I(i)_\alpha(p) = \begin{cases} p' & \text{if } p = p'\alpha, \text{ i.e., } p \text{ starts with } \alpha, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, $I(i)_i \cong k$ with basis $\{e_i\}$ (the trivial path at i).

Define

$$\Phi\colon \operatorname{Hom}(M, I(i)) \longrightarrow M_i^* = \operatorname{Hom}_k(M_i, k)$$

by $\Phi(f) = f_i$, where we identify $I(i)_i \cong k$ via $e_i \mapsto 1$.

Φ is injective. Suppose $f\colon M \to I(i)$ is a morphism with $f_i = 0$. Let $j \in Q_0$ and let $m \in M_j$. We must show $f_j(m) = 0$, i.e., that for every path $p\colon j \to i$, the coefficient of p in $f_j(m)$ is zero. Write $p = \alpha_r \cdots \alpha_1$ where $\alpha_k\colon j_{k-1} \to j_k$ with $j_0 = j$ and $j_r = i$. Since f is a morphism, commutativity gives

$$f_i \circ \psi_{\alpha_r} \circ \cdots \circ \psi_{\alpha_1} = I(i)_{\alpha_r} \circ \cdots \circ I(i)_{\alpha_1} \circ f_j.$$

Under the identification $I(i)_i \cong k$, the composition $I(i)_{\alpha_r} \circ \cdots \circ I(i)_{\alpha_1}$ applied to $f_j(m)$ extracts exactly the coefficient of p in $f_j(m)$. Since $f_i = 0$, the left-hand side vanishes, so every such coefficient is zero. Hence $f_j = 0$ for all j, and Φ is injective.

Φ is surjective. Let $\lambda \in M_i^* = \operatorname{Hom}_k(M_i, k)$. We construct a morphism $f^\lambda\colon M \to I(i)$ with $f_i^\lambda = \lambda$. For each vertex j and each $m \in M_j$, define

$$f_j^\lambda(m) = \sum_{p\colon j \to i} \lambda\big(\psi_p(m)\big) \cdot p,$$

where $\psi_p = \psi_{\alpha_r} \circ \cdots \circ \psi_{\alpha_1}$ denotes the composition of the structure maps of M along the path $p = \alpha_r \cdots \alpha_1$. This is a finite sum since Q is acyclic. In particular, $f_i^\lambda(m) = \lambda(m) \cdot e_i$, which corresponds to λ under the identification $I(i)_i \cong k$.

We verify that f^λ is a morphism. Let $\alpha\colon j \to \ell$ be an arrow. For $m \in M_j$:

$$f_\ell^\lambda(\psi_\alpha(m)) = \sum_{q\colon \ell \to i} \lambda\big(\psi_q(\psi_\alpha(m))\big) \cdot q = \sum_{q\colon \ell \to i} \lambda\big(\psi_{q\alpha}(m)\big) \cdot q.$$

On the other hand,

$$I(i)_\alpha(f_j^\lambda(m)) = \sum_{p\colon j \to i} \lambda\big(\psi_p(m)\big) \cdot I(i)_\alpha(p).$$

A path $p\colon j \to i$ satisfies $I(i)_\alpha(p) \neq 0$ if and only if p starts with α, i.e., $p = q\alpha$ for some path $q\colon \ell \to i$. In that case $I(i)_\alpha(q\alpha) = q$. Hence

$$I(i)_\alpha(f_j^\lambda(m)) = \sum_{q\colon \ell \to i} \lambda\big(\psi_{q\alpha}(m)\big) \cdot q = f_\ell^\lambda(\psi_\alpha(m)).$$

Thus f^λ is a morphism and $\Phi(f^\lambda) = \lambda$, so Φ is surjective.

Conclusion. Since all vector spaces are finite-dimensional, $M_i^* \cong M_i$ as vector spaces (non-canonically). Therefore

$$\operatorname{Hom}(M, I(i)) \cong M_i^* \cong M_i$$

as k-vector spaces. $\square$

8. Let i and j be vertices in the quiver Q. Prove that the vector space $\mathrm{Hom}(I(i), I(j))$ has a basis consisting of all paths from j to i in Q. In particular, $\mathrm{End}(I(i)) \simeq k$.

Proof. By Problem 7 above, for any representation X and any vertex j, the isomorphism

$$\mathrm{Hom}(X, I(j)) \cong X_j^*$$

holds (where $X_j^* = \mathrm{Hom}_k(X_j, k) \cong X_j$ as a vector space in finite dimensions). Applying this with $X = I(i)$, we obtain

$$\mathrm{Hom}(I(i), I(j)) \cong I(i)_j^*.$$

By definition, $I(i)_j$ is the k-vector space with basis the set of all paths from j to i in Q. Hence

$$\dim \mathrm{Hom}(I(i), I(j)) = \dim I(i)_j = |\{\text{paths from } j \text{ to } i\}|.$$

We now exhibit an explicit basis. For each path $p\colon j \to i$, we construct a morphism $\theta^p\colon I(i) \to I(j)$. By the isomorphism $\mathrm{Hom}(I(i), I(j)) \cong I(i)_j^*$, a morphism $f\colon I(i) \to I(j)$ corresponds to a linear functional on $I(i)_j$. The path $p \in I(i)_j$ is a basis element, and we let $p^* \in I(i)_j^*$ be the corresponding dual basis element (the functional satisfying $p^*(p) = 1$ and $p^*(q) = 0$ for any other basis path $q \neq p$ from j to i). Let θ^p be the morphism corresponding to p^*.

Explicitly, for a vertex ℓ and a basis element $c \in I(i)_\ell$ (a path $c\colon \ell \to i$), the morphism θ^p is given by

$$\theta^p_\ell(c) = \begin{cases} c' & \text{if } c = pc', \text{ i.e., the path } c \text{ factors as } \ell \xrightarrow{c'} j \xrightarrow{p} i, \\ 0 & \text{otherwise.} \end{cases}$$

Here, $c'\colon \ell \to j$ is the remaining path after removing the suffix p. We verify this is a morphism. Let $\alpha\colon \ell \to \ell'$ be an arrow. For a basis element $c\colon \ell \to i$ of $I(i)_\ell$:

- The map $I(i)_\alpha(c)$ is nonzero only if c starts with α, say $c = c''\alpha$. In that case $I(i)_\alpha(c) = c'' \in I(i)_{\ell'}$.
- We need $\theta^p_{\ell'}(I(i)_\alpha(c)) = I(j)_\alpha(\theta^p_\ell(c))$.

If $c = pc'$ for some path $c'\colon \ell \to j$, and also c starts with α (so c' starts with α, say $c' = c''\alpha$), then $I(i)_\alpha(c) = pc''$ and $\theta^p_{\ell'}(pc'') = c''$. On the other hand, $\theta^p_\ell(c) = c'$ and $I(j)_\alpha(c') = c''$ (since $c' = c''\alpha$ starts with α). Both sides agree. The other cases are similarly verified.

Since the paths p from j to i form a basis of $I(i)_j$, the dual functionals $\{p^*\}$ form a basis of $I(i)_j^*$, and thus the morphisms $\{\theta^p\}$ form a basis of $\mathrm{Hom}(I(i), I(j))$.

The special case $i = j$. The only path from i to i is the trivial path e_i (since Q has no oriented cycles). Therefore

$$\mathrm{End}(I(i)) = \mathrm{Hom}(I(i), I(i)) \cong k,$$

with basis $\{\theta^{e_i}\} = \{\mathrm{id}_{I(i)}\}$. $\square$